

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (EC) Examination****November 2014****EC310.01 Optical Communication****Date: 22.11.2014, Saturday****Time: 01:30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the question below.****[07]**

- a. Define channel capacity.
- b. What is refractive index?
- c. State the difference between reflection and refraction.
- d. Light traveling in optical fiber follows which of the following principles.
 - a) Huygen's principle
 - b) Reflection theory
 - c) Light theory
 - d) Snell's law
- e. The speed of light in plastic compared to the speed of light in air is
 - a) Slower
 - b) Faster
 - c) The same
 - d) Either lower or faster
- f. What are skew rays?
- g. Define the internal quantum efficiency.

Q – 2.a Consider the multimode silica fiber that has a core refractive index $n_1 = 1.48$ and a cladding index $n_2 = 1.46$. Find the Critical angle and Acceptance angle in air.

[04]

Q – 2.b Answer any two questions.

[10]

- 1) Explain conditions for the phenomenon of total internal reflection.
- 2) Explain the key elements of optical fiber communication system.
- 3) List the different attenuation mechanisms in fiber. Explain any one in detail.

Q - 3 Answer the following.

- a.** Explain the types of losses in optical fiber in detail. [07]
- b.** 1) Explain Surface emitting LED with neat diagram. [04]
2) Explain what is meant by equilibrium NA. [03]

OR

- 1) Discuss the most common type of mechanical misalignment occurring between two joined fibers and its effect. [04]
- 2) Explain three key transition processes involved in laser action. [03]

SECTION – II**Q - 4 Answer the question below.****[07]**

- a.** SONET stands for
- System Optical Network
 - Synchronous Optical Network
 - Silica Optic Network
 - System Optical Fiber Net
- b.** Define Optical coupler.
- c.** Which optical detector is used when high sensitivity and bandwidth are required?
- APD
 - PIN
 - Photo detector
- d.** Which of the following is determined by an OTDR?
- Distance to trouble
 - Length of fiber
 - Refractive index
 - All of these
- e.** Photodiodes operate properly with ____.
- Forward bias
 - Reverse bias
 - Neither A or B
- f.** Define carrier multiplication.
- g.** Define splitting ratio in case of optical coupler.

Q – 5.a A given Silicon avalanche photodiode has a quantum efficiency of 65% at a wavelength of 900 nm. Suppose 0.5 μ W of optical power produce a multiplied photocurrent of 10 μ A. Then find out the multiplication factor M. ($q = 1.6 \times 10^{-19}$ C , $h = 6.625 \times 10^{-34}$ Js , $c = 3 \times 10^8$ m/s) **[05]**

OR

Q – 5.a In One optical communication link have the photo diode which required receiver input signal of -42dbm and optical source that can transmit -13dbm average optical power level in to the fiber fly head with 50 μ m core diameter. Assume that 1db loss occur when the fiber fly head is connected to the cable and another 1db connector loss. Link also includes the 6db system margin. Find the possible transmission distant with $\alpha_f = 3.5$ db/km **[05]**

Q – 5.b Explain principle of operation, the structure and the characteristics of the pin diode. **[04]**

Q – 5.c Compare silicon RAPD with PIN photo detector with respect to the structure, principle of operation and the performance. State the advantages of RAPD. **[05]**

OR

Q – 5.c List out different types of photo detectors and compare their performance. **[05]**

Q – 6.a Define **[02]**

- 1) Link Power Budget
- 2) Rise Time Budget

Q – 6.b Explain SONET frame structure. **[07]**

Q – 6.c What are the key system features of WDM? **[03]**

Q – 6.d Draw schematic diagram of a typical optical receiver. **[02]**

OR

Q – 6.a Explain 2 x 2 fiber coupler. What are the different losses associated with it? **[06]**

Q – 6.b What is Switching? Give the Difference Between Packet switching and Burst Switching. **[04]**

Q – 6.c Explain digital signal transmission in optical communication with figure. **[04]**
